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Molecular Characterization of Native and Recombinant Apolipoprotein A-I_{Milano} Dimer

THE INTRODUCTION OF AN INTERCHAIN DISULFIDE BRIDGE REMARKABLY ALTERS THE PHYSICOCHEMICAL PROPERTIES OF APOLIPOPROTEIN A-I*

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The disulfide-linked dimer of apolipoprotein A-I_{Milano} (A-I_M/A-I_M), a natural Arg^{173} → Cys variant of apoA-I, was purified from carriers' plasma and produced in *Escherichia coli*. The recombinant A-I_M/A-I_M is identical to native A-I_M/A-I_M, by mass spectrometry, SDS-polyacrylamide gel electrophoresis, and isoelectric focusing. Lipid-free A-I_M/A-I_M undergoes concentration-dependent self-association similar to apoA-I, but at all concentrations apoA-I is more self-associated than A-I_M/A-I_M. Far-ultraviolet CD spectra of A-I_M/A-I_M reveal a highly α-helical structure predicted to be ~65% in the lipid-free and ~78% in the lipid-associated states, *versus* 43 and 73% for apoA-I. A significant loss of α-helix occurs below pH 3.5 and above pH 10 in both apoA-I and A-I_M/A-I_M; A-I_M/A-I_M constantly shows a higher α-helical content than apoA-I over the entire pH range (1.7–12.8), suggesting that hydrophobic forces stabilize the interaction between the two A-I_M chains. Indeed, and differently from apoA-I, the α-helical content of A-I_M/A-I_M is minimally affected by solvent ionic strength. The aromatic side chains in both lipid-free and lipid-bound A-I_M/A-I_M are immobilized in a more asymmetric and hydrophobic environment than in lipid-free apoA-I, the conformation of A-I_M/A-I_M being instead similar to that achieved by apoA-I following interaction with lipids. The present findings prove that rA-I_M/A-I_M is structurally identical to the native protein; the conformation of A-I_M/A-I_M is remarkably different from that of apoA-I, thus possibly explaining some of the peculiar functional properties of the apoA-I_{Milano} dimer.

High density lipoproteins (HDL)[1] and its major protein component, apolipoprotein A-I (apoA-I), play crucial roles in cholesterol homeostasis, primarily acting as a vehicle of cholesterol in the "reverse cholesterol transport," the process by which excess cholesterol is removed from peripheral tissues and transported to the liver for final elimination (1).

ApoA-I actively modulates the reverse cholesterol transport by acting at various stages in the process. Reconstituted HDL, or naturally occurring pre-β-migrating HDL, containing apoA-I as the only protein component, are more efficient acceptors of cell cholesterol than those containing other exchangeable apolipoproteins (2, 3), and lipid-free apoA-I can mediate cellular cholesterol efflux by forming pre-β-HDL-like particles in the extracellular space (4). Esterification of cell-derived cholesterol, the second step in reverse transport, is also remarkably dependent on the ability of apoA-I to activate the lecithin:cholesterol acyltransferase enzyme (5). Finally, HDL can directly transport the lecithin:cholesterol acyltransferase-derived cholesteryl esters to the liver, in a process possibly mediated by recognition of apoA-I (6).

ApoA-I is a single polypeptide chain, composed of 243 amino acids (7). Studies of apoA-I structure have mainly focused on secondary structure because of the high percentage of α-helix and the motif of amphipathic helices, which is believed to mediate protein-lipid interaction in the lipoproteins (8). The most striking feature of the apoA-I molecule is the presence of internal repeat units of 11 or 22 amino acids, with the periodicity of an amphipathic α-helix (9). This structure allows for the main biological activities of apoA-I, *i.e.* lipid-binding (8), lecithin:cholesterol acyltransferase activation (5), stimulation of cholesterol efflux from lipid-loaded cells (10), and interaction with the putative HDL receptor (11). The amino-terminal end of the protein has significantly less α-helix than the COOH-terminal fragment (residues 149–243), which is organized in highly packed antiparallel amphipathic helices, and has the highest affinity for lipids (12, 13). Less is known about the tertiary structure of apoA-I; a structural model has been recently proposed, with a domain of interacting α-helical segments in the carboxyl terminus and a globular domain in the amino terminus (13–15). Physicochemical (14, 16), monoclonal antibody (17, 18), and limited proteolysis studies (13) suggest that a hinge domain may exist in the apoA-I molecule (9), possibly involved in lecithin:cholesterol acyltransferase activation (5, 19, 20) and promotion of cell cholesterol efflux (21).

Apolipoprotein A-I_{Milano} (A-I_M) is the first described molecular variant of human apolipoproteins (22) and seems to confer to the carriers an increased protection against vascular disease (23). ApoA-I_M is characterized by an Arg^{173} → Cys substitution (24), resulting in the formation of disulfide-linked homodimers (A-I_M/A-I_M) and heterodimers with apoA-II. The presence of these covalently linked apoA-I molecules is likely responsible for most of the structural/metabolic abnormalities of carriers'

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[1] The abbreviations used are: HDL, high density lipoprotein; apoA-I, apolipoprotein A-I; A-I_M, A-I_{Milano}; A-I_M/A-I_M, A-I_{Milano} homodimer; rA-I_M/A-I_M, *recombinant* A-I_M/A-I_M; GdnHCl, guanidine hydrochloride; PAGE, polyacrylamide gel electrophoresis; DMPC, dimyristoylphosphatidylcholine; DMS, dimethylsuberimidate; HPLC, high performance liquid chromatography; Tricine, *N*-[2-hydroxy-1,1-bis(hydroxymethyl)ethyl]-glycine.

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HDL. Indeed, in an *in vitro* system, the interconversion of small, A-I$_M$/A-I$_M$-containing HDL is impaired (25), while *in vivo* the A-I$_M$ dimers are catabolized more slowly than monomeric A-I$_M$ or normal apoA-I (26).

The Arg^{173} → Cys replacement in monomeric apoA-I$_M$ significantly alters the structural and lipid-binding properties of apoA-I (27). To identify alterations induced in apoA-I structure by the introduction of an interchain disulfide bond, the present study compared the spectroscopic properties of A-I$_M$/A-I$_M$ and apoA-I. A-I$_M$/A-I$_M$ was either purified from carriers' plasma or expressed in *Escherichia coli*; the recombinant A-I$_M$/A-I$_M$ (rA-I$_M$/A-I$_M$) proved to be structurally identical to native A-I$_M$/A-I$_M$, both showing physicochemical properties distinct from that of normal apoA-I.

EXPERIMENTAL PROCEDURES

Purification of Plasma A-I_M/A-I_M—HDL (d = 1.063–1.21 g/ml) were isolated from A-I$_M$ plasma by sequential ultracentrifugation in a 50.2 Ti rotor (Beckman) and washed at 40,000 rpm, 4 °C for 48 h. HDL were extensively dialyzed against 5 mM NH_4HCO_3, containing 0.01% EDTA, pH 7.4, lyophilized, and delipidated with diethylether/ethanol (3/1, v/v). HDL apolipoproteins were then solubilized in 0.1 M Tris-HCl, 0.04% EDTA, 0.01% NaN_3, pH 7.4, containing 6 M guanidine hydrochloride (GdnHCl), and applied to a Sephacryl S-300 HR column (2.6 × 300 cm) (Pharmacia Biotech Inc.) equilibrated with 0.1 M Tris-HCl, 0.04% EDTA, 0.01% NaN_3, pH 7.4, containing 4 M GdnHCl. Fractions corresponding to A-I$_M$/A-I$_M$ were concentrated and reapplied to the same column. The purified A-I$_M$/A-I$_M$ was dialyzed against 5 mM NH_4HCO_3, 0.01% Na_2-EDTA, pH 7.4, and lyophilized. The purity (>95%) of A-I$_M$/A-I$_M$ preparations was confirmed by polyacrylamide gel electrophoresis (PAGE) in 0.5% sodium dodecyl sulfate, using Coomassie protein staining. Aliquots of A-I$_M$/A-I$_M$ were stored at −80 °C in lyophilized form. Before use, A-I$_M$/A-I$_M$ was dissolved in 20 mM phosphate buffer, pH 7.4, containing 6 M GdnHCl and extensively dialyzed against 20 mM phosphate buffer, pH 7.4. ApoA-I was purified from normal human plasma, essentially as described previously (27).

Expression and Purification of rA-I_M/A-I_M—rA-I$_M$/A-I$_M$ was expressed in the *E. coli* K12 derivative strain BC 0050 (*xyl-7*, *ara-14*, T4^{R2}) and excreted to the growth medium. The expression vector was derived form pTrc99A (28). The A-I$_M$ cDNA was prepared by site-directed mutagenesis of apoA-I cDNA (29) and placed immediately downstream of an *OmpA* signal sequence derivative, where the second to last amino acid was changed from Gln to Asn. Two transcription terminators were placed downstream of the A-I$_M$ gene and the ampicillin resistance marker of pTrc99A was replaced with a kanamycin resistance marker.

Fermentation was performed at 30 °C and pH 7.0 in a fed-batch bioreactor of 75 liters as in Forsberg *et al.* (30) with the following exceptions. The medium contained 5 g/liter yeast extract and 0.05 g/liter kanamycin. The culture was grown until the absorbance at 600 nm was 0.6. A-I$_M$ synthesis was then induced by adding 0.5 mM isopropyl-1-thio-β-D-galactopyranoside, and the temperature was raised to 37 °C. Six hours after induction, the concentration of A-I$_M$ in the supernatant, determined by radioimmunoassay (Apolipoprotein A-I RIA, Pharmacia Biotech Inc.), was ~2 g/liter.

Cells were separated by centrifugation at 3,000 rpm for 20 min, and the medium was diluted with distilled water to a conductivity below 10 mS/cm prior to loading onto a Q-Sepharose FF (Pharmacia Biotech Inc.) column, equilibrated with 0.02 M sodium phosphate, pH 8.0. The column was washed and then eluted with 0.4 M NaCl in the same buffer. A-I$_M$-containing fractions were applied to a phenyl-Sepharose FF gel column, equilibrated with 0.02 M phosphate buffer, pH 7.5, containing 0.75 M ammonium sulfate. After washing with 0.02 M phosphate buffer, pH 7.5, containing 0.04 M ammonium sulfate, 0.02 M phosphate, A-I$_M$ was eluted with distilled water, and aliquots of this material were loaded onto a Superdex 200 BPG 100 column, equilibrated with 5 mM NH_4HCO_3, 30% 1-propanol and eluted at a flow rate of 25 ml/min. Fractions containing rA-I$_M$/A-I$_M$ were pooled, and the buffer was changed to 20 mM sodium phosphate, pH 8.0, by diafiltration on Omega 8k filter cassettes. The concentrated rA-I$_M$/A-I$_M$ pool was applied to a Q-Sepharose FF column, equilibrated with 0.02 M sodium phosphate, pH 7.5, containing 6 M urea and was eluted with a linear gradient of sodium chloride (0–0.1 M). Pooled fractions were concentrated and diafiltrated against 0.2 M phosphate buffer, pH 7.8, containing 1% mannitol and lyophilized. Before use, rA-I$_M$/A-I$_M$ was dissolved in 20 mM phosphate buffer, pH 7.4, containing 6 M GdnHCl and extensively dialyzed against 20 mM phosphate buffer, pH 7.4.

Characterization of rA-I_M/A-I_M—rA-I$_M$/A-I$_M$ (250 μg) was digested with endoproteinase Lys-C (Boehringer Mannheim) (2 μg in 50 μl of water) for 18 h at 37 °C in 200 μl of 25 mM Tris-HCl, pH 8.5, 1 mM EDTA. The resulting peptides were separated by reverse-phase HPLC, using a Brownlee Aquapore butyl column developed at 40 °C with a gradient of 5–40% acetonitrile in water, containing 0.25% pentafluoropropionic acid. Each peptide was sequenced by automated Edman degradation, and the molecular mass determined by a BioIon 20, ^{252}Cf plasma desorption time-of-flight mass spectrometer (Applied Biosystems). A 59-residue COOH-terminal fragment (H2: residues 185–243) was prepared by cleavage with hydroxylamine. Lyophilized rA-I$_M$/A-I$_M$ (480 μg) was dissolved in 1 ml of cleavage solution containing 2 M hydroxylamine, 3 M GdnHCl, 0.2 M NaOH, and 2 mM EDTA, pH 9.4, and incubated for 5 h at 40 °C. The COOH-terminal fragment was purified by reverse-phase HPLC, using a YMC-Pak Protein column, eluted with a gradient of 10–60% acetonitrile in water, containing 0.25% pentafluoropropionic acid.

For determination of COOH-terminal amino acids, rA-I$_M$/A-I$_M$ or its COOH-terminal H2 fragment (4 nmol) were dissolved in 85 μl of 0.21 M acetic acid, containing 0.05% Brij 35 and 6 nmol of norleucine as an internal standard. Carboxypeptidase P (Sigma) (15 μg in 15 μl of water) was added, and the solution was incubated at 20 °C. Samples (25 μl) were withdrawn after 0, 1, 2, and 4 min, and the digestion was stopped by adding 10 μl of 50% trifluoroacetic acid. The released amino acids were analyzed by the Picotag method (Waters). The NH_2-terminal amino acid sequence was determined using a Hewlett-Packard G1000A protein sequencer or a Milligen Biosearch ProSequencer type 6600, with Sequelon AA membranes (Millipore/Milligen).

The molecular mass of the intact protein was determined using positive electrospray mass spectrometry (VG Quattro, Fisons Instruments). The samples were applied in 50% methanol, 1% acetic acid at a flow rate of 5 μl/min.

Amino Acid Analysis and Protein Concentration—Protein concentration of stock solutions was assayed by amino acid analysis, performed on a Beckman 6300 amino acid analyzer after acidic hydrolysis of samples in 6 M HCl for 45 min at 155 °C.

Spectroscopy—Circular dichroism (CD) spectra were recorded with a Jasco J500A spectropolarimeter at the constant temperature of 25 °C. Cells were 0.01 and 0.1 cm for the region below 250 nm, and 1.0 cm above 250 nm. Molar mean residue ellipticity (θ) was expressed in degrees·cm^2·dmol^{-1}, and calculated as:

$$[\theta] = \theta_{obs} \cdot 115/10 \cdot l \cdot c \qquad \text{(Eq. 1)}$$

where θ_{obs} is the observed ellipticity in degrees, 115 is the mean residue molecular weight of the proteins, l is the optical path length in centimeters, and c the protein concentration in grams/ml. All of the spectra were base line-corrected. The α-helical content was calculated by the method of Chang *et al.* (31).

The ultraviolet absorption spectra were measured with a Jasco Uvidec-610 spectrophotometer equipped with on-line digital data processor. Differential absorption spectra were calculated by subtraction of the base-line spectrum stored in memory. Topographical location of tyrosyl residues was investigated by second-derivative analysis, according to Ragone *et al.* (32). The a and b values were calculated as the peak to trough distances between the minima at 283 and 290.5 nm, and the maxima at 287 and 295 nm, respectively. The fractional tyrosine exposure was then expressed as $\alpha = (r - r_a)/(r_u - r_a)$, where r is the ratio a/b, measured from the second-derivative absorption spectra, r_u is the ratio for the protein dissolved in 6 M GdnHCl, and r_a is the ratio calculated for a model protein solution containing the same tyrosine/tryptophan ratio dissolved in ethylene glycol to simulate residues completely buried in the protein interior. For apoA-I, r_u and r_a were 1.20 and 0.35, respectively.

Reconstitution Procedures—Weighed amounts of dimyristoylphosphatidylcholine (DMPC) were dissolved in chloroform and the solvent evaporated under N_2; any remaining solvent was removed under vacuum for 2 h. Dispersions of DMPC in 10 mM Tris-HCl, 1 mM EDTA, pH 7.4, containing 8.5% KBr, were mixed with apoA-I, A-I$_M$/A-I$_M$, or rA-I$_M$/A-I$_M$ at 2.5:1, DMPC:protein weight ratio and incubated for 20 h at 25 °C (27). The complexes were stored at 4 °C and dialyzed against 20 mM phosphate buffer, pH 7.4, before use.

Electrophoresis—SDS-PAGE was performed on 10% acrylamide gels using the Tris-Tricine buffer system of Schagger and Van Jagou (33) in

[2] G. Palm, H. Wadensten, A. Hammarström, A. Olsson, A. Karlström, and T. Sejlitz, unpublished results.

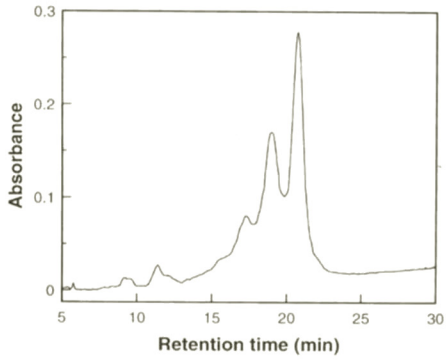


FIG. 1. **Reverse-phase HPLC of rA-I$_M$/A-I$_M$.** The purified rA-I$_M$/A-I$_M$ was applied to a YMC-Pak Protein reverse phase column and eluted with a water-acetonitrile gradient, containing 0.25% pentafluoropropionic acid. Three major peaks were separated; the material eluting at ~21 min corresponds to the full-length product (see text) and has been used in further experiments.

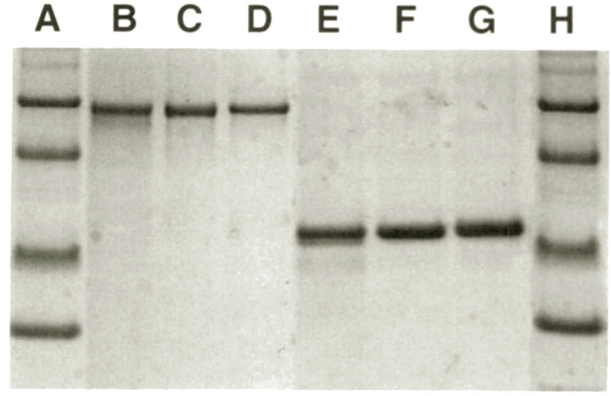


FIG. 2. **SDS-PAGE of native and recombinant A-I$_M$/A-I$_M$, in absence and presence of β-mercaptoethanol.** *Lanes A* and *H*, molecular weight markers; *lanes B* and *E*, rA-I$_M$/A-I$_M$; *lanes C* and *F*, HPLC-purified rA-I$_M$/A-I$_M$; *lanes D* and *G*, native A-I$_M$/A-I$_M$. Samples in lanes E–G were reduced with 2% β-mercaptoethanol.

a Mini Protean slab minigel apparatus (Bio-Rad). In selected samples, reduction of disulfide bonds was accomplished by addition of dithiothreitol to the sample buffer, to give a final concentration of 84 mM. The gels were scanned at 627 nm using an UltroScan XL laser densitometer (Pharmacia Biotech Inc.). Two-dimensional charge-size separations were achieved with a nonlinear 4–10 immobilized pH gradient in the first dimension and a 7.5–17.5% polyacrylamide gradient in the SDS-PAGE (34). Cross-linking was performed by the addition of 1 part dimethylsuberimidate (DMS) solution, 10 mg/ml in 0.3 M triethanolamine, to 10 parts protein solution (35). Cross-linked samples were analyzed by SDS-PAGE on 4–10% acrylamide gradient slab gels.

RESULTS AND DISCUSSION

Characterization of rA-I$_M$/A-I$_M$—The HPLC profile of purified rA-I$_M$/A-I$_M$ showed three major peaks (Fig. 1). Mass spectrometry analysis revealed that the peak eluting at ~21 min corresponds to the full-length product, while the other two peaks are disulfide-linked A-I$_M$ dimers containing one (peak at 19 min) or two (peak at 17 min) NH_2-terminally truncated monomers (see below). The full-length protein was further purified by reverse-phase HPLC, eliminating most of the truncated monomers (Fig. 2, *lane C*); this product was used in following experiments. The rA-I$_M$/A-I$_M$ co-migrated with the native protein on SDS-PAGE in both non-reduced and reduced conditions (Fig. 2). No protein thiols were detected by the Ellman's (36) assay in preparations of either native or recombinant A-I$_M$/A-I$_M$, indicating that the purified proteins were entirely in the disulfide form. Two-dimensional gel electrophoretic separation of apoA-I revealed three major and three minor isoproteins (Fig. 3) (37). Reduced rA-I$_M$/A-I$_M$, as well as native A-I$_M$ (38), showed the same isoproteins, but with one less positive charge compared with normal apoA-I, as expected from the amino acid substitution in the mutant apolipoprotein.

The amino acid composition of both A-I$_M$/A-I$_M$ and rA-I$_M$/A-I$_M$ was in close agreement with that deduced from the DNA sequence (39). The primary structure of rA-I$_M$, as well as the sequencing strategy, is presented in Fig. 4. Determination of the NH_2-terminal sequence showed, besides the correct translation product, shorter NH_2-terminal sequences in preparation containing truncated rA-I$_M$. The molecular mass (6,530 ± 3 Da), amino acid composition and sequence of the COOH-terminal H2 fragment, together with the direct determination of the COOH-terminal Gln by carboxypeptidase P, confirmed that the COOH terminus of rA-I$_M$ was correct. The rA-I$_M$ primary sequence was deduced from direct sequencing of the peptides generated by specific cleavage with lysyl and aspartyl endoproteinases. Their molecular mass deviated from theoretical values by less than 0.2%. The direct estimation of the rA-I$_M$/A-I$_M$ mass by electrospray ionization mass spectrometry gave a value of 56,053 ± 28 Da, *i.e.* in good agreement with the calculated molecular mass deduced from the amino acid sequence (56,049 Da). This finding argues against the occurrence of post-

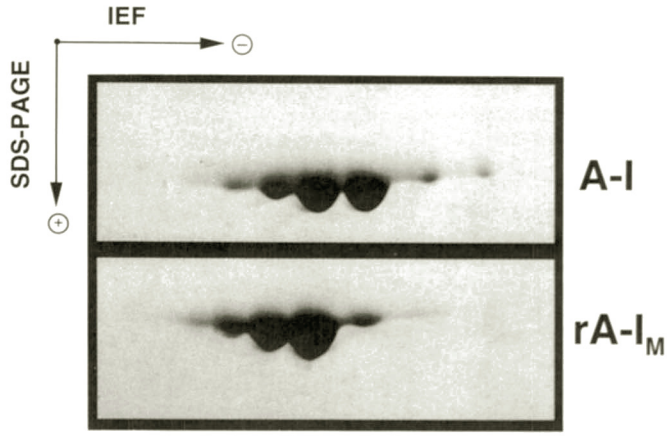


FIG. 3. **Two-dimensional gel electrophoresis of apoA-I and reduced A-I$_M$/A-I$_M$.** Isoelectric focusing on immobilized pH gradient was performed in the first dimension, followed by SDS-PAGE in the second dimension. Only the portions of the gels containing the protein isoforms are shown.

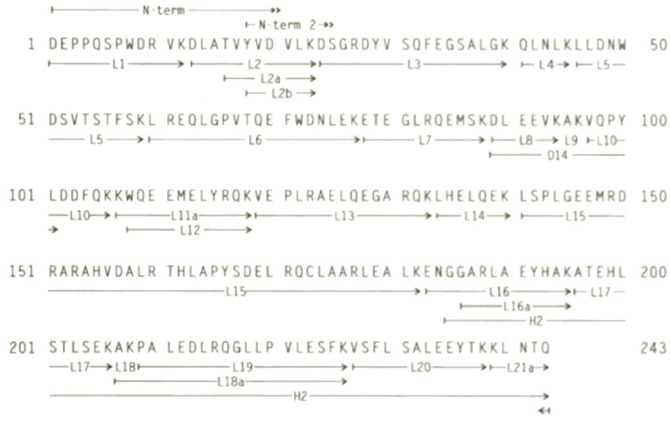
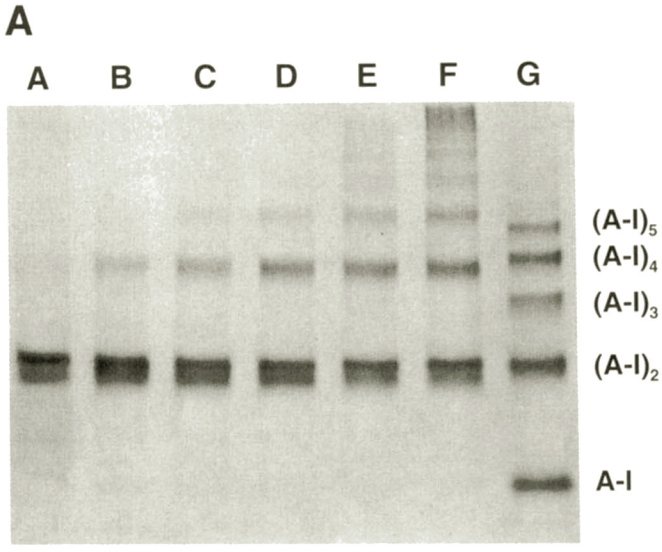


FIG. 4. **Amino acid sequence of rA-I$_M$ and sequencing strategy.** The *arrows above* the sequence refer to the automatic Edman degradation performed on full-length rA-I$_M$ (*N-term*) and on the truncated protein (*N-term 2*). Peptides generated by endoproteinase Lys-C are designated *L1–L21a*. Peptide D14 was obtained by endoproteinase *Asp*N digestion. The COOH-terminal H2 fragment was prepared by cleavage with hydroxylamine. All peptides were sequenced in their entirety. The small arrow (←) refers to the COOH-terminal residue identified after carboxypeptidase digestion.



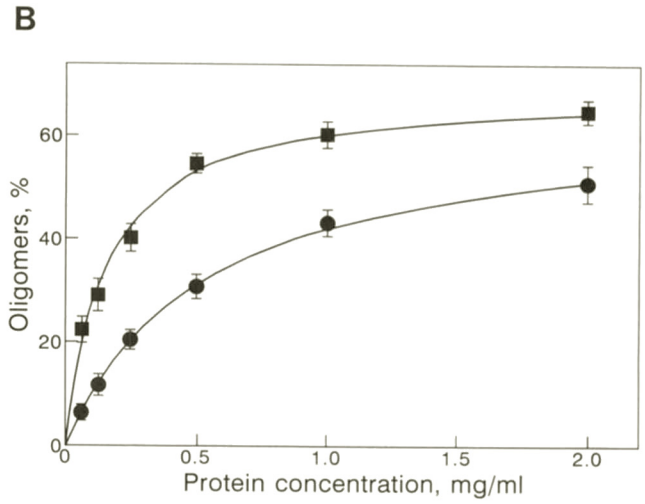


FIG. 5. **Cross-linking of rA-I$_M$/A-I$_M$.** *Panel A*, SDS-PAGE of cross-linked rA-I$_M$/A-I$_M$. Equal amounts of rA-I$_M$/A-I$_M$ were diluted with water and cross-linked with DMS before separation on a 4–10% acrylamide gradient gel. Cross-linked apoA-I was used as molecular weight standard (*lane G*). The concentrations of rA-I$_M$/A-I$_M$ present in the reaction mixture were: *A*, 0.0625 mg/ml; *B*, 0.125 mg/ml; *C*, 0.25 mg/ml; *D*, 0.5 mg/ml; *E*, 1.0 mg/ml; *F* and *G*, 2.0 mg/ml. *Panel B*, concentration dependence of the self-association of apoA-I (*squares*) and rA-I$_M$/A-I$_M$ (*circles*). Proteins were cross-linked at various concentrations and electrophoresed on gradient gels; the percentage of oligomers was calculated from the SDS-PAGE scans and reported as a function of protein concentration in the reaction mixture.

translational modifications in rA-I$_M$/A-I$_M$.

Cross-linking Studies—Cross-linking with DMS has been repeatedly used to evaluate the self-association of soluble apolipoproteins (35, 40). Cross-linking of apoA-I gives five prominent bands, identified as monomer (M_r 28,000) through pentamer (M_r 140,000) (35). The cross-linking pattern is sensitive to protein concentration, monomers and dimers predominating at the lower values (<0.5 mg/ml), and tetramers and pentamers becoming major species at higher concentrations (35). Cross-linking experiments with both native and recombinant A-I$_M$/A-I$_M$ gave superimposable patterns and showed a similar formation of high molecular weight forms (Fig. 5*A*). Cross-linked A-I$_M$/A-I$_M$ gave four prominent bands; some high molecular weight material was seen at high protein concentration, but was negligible compared to the total area of the prominent bands (Fig. 5*A*). A-I$_M$/A-I$_M$ co-migrated, as expected, with

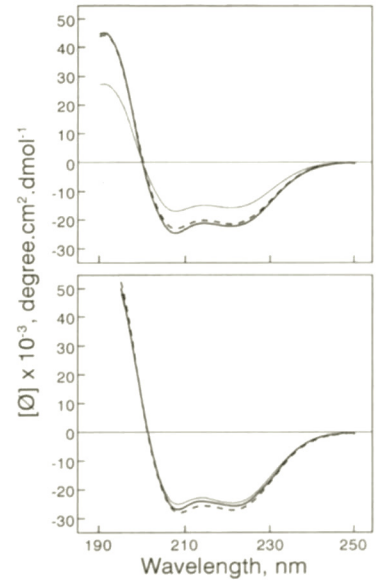


FIG. 6. **Far-ultraviolet circular dichroism spectra of A-I$_M$/A-I$_M$ (*continuous lines*), rA-I$_M$/A-I$_M$ (*dashed lines*), and apoA-I (*thin continuous lines*), in the lipid-free (*top panel*) and lipid-associated (*bottom panel*) states.** Protein samples (0.1 mg/ml) were equilibrated in 20 mM phosphate buffer, pH 7.4, before spectra were recorded at 25 °C.

the apoA-I dimer (M_r 56,000). The band co-migrating with the apoA-I tetramer (M_r 112,000) was identified as a cross-linked species made of two A-I$_M$/A-I$_M$ molecules. Notably, no bands co-migrating with the apoA-I trimer and pentamer were present in cross-linked A-I$_M$/A-I$_M$, while the two high molecular weight bands were tentatively identified as cross-linked species containing three and four A-I$_M$/A-I$_M$ molecules. As previously shown for apoA-I (35), the A-I$_M$/A-I$_M$ cross-linking pattern was sensitive to protein concentration (Fig. 5*A*). The percentage of apoA-I and A-I$_M$/A-I$_M$ oligomers, calculated from the SDS-PAGE gels assuming that the amount of dye bound per unit weight is constant for all species, is plotted as function of protein concentration in Fig. 5*B*. At all concentrations, the extent of self-association was significantly lower for A-I$_M$/A-I$_M$ than apoA-I. Some lipid-binding protein (41–43), which exist as up and down α-helical bundles (44, 45), do not self-associate in solution and remain monomeric up to very high protein concentrations. Thus, it seems reasonable to speculate that A-I$_M$/A-I$_M$ may have a molecular architecture intermediate between those of self-associating and monomeric globular apolipoproteins.

Circular Dichroism Studies—The secondary and tertiary structures of native A-I$_M$/A-I$_M$ and rA-I$_M$/A-I$_M$, in their lipid-free and lipid-associated states, were examined by CD spectroscopy. A protein concentration of 0.1 mg/ml, at which both apoA-I (35, 46) and A-I$_M$/A-I$_M$ (Fig. 5) are essentially monomeric, was initially used. In the far-ultraviolet, rA-I$_M$/A-I$_M$ and the native protein gave superimposable CD spectra (Fig. 6). The CD spectra are indicative of a highly α-helical structure, exhibiting negative troughs at 208 and 222 nm, together with a positive band at 190–192 nm. ApoA-I gave a similar spectrum, but the values of negative ellipticity for the troughs at 208 and 222 nm were lower than in A-I$_M$/A-I$_M$ (Fig. 6, Table I). The analysis of the CD spectra by the method of Chang *et al.* (31) reveals an α-helical content of 64% in rA-I$_M$/A-I$_M$, 66% in native A-I$_M$/A-I$_M$, and 43% in apoA-I, this latter being close to that previously reported by us and others (27, 47). All proteins approached a maximum molar ellipticity value (Table I) upon addition of the helix-inducing solvent trifluoroethanol (50%), the calculated

TABLE I
Molar ellipticities and calculated α-helical contents

CD spectra were recorded between 190 and 250 nm, at 25 °C, and the α-helical content was calculated by the method of Chang *et al.* (31). Proteins (0.1 mg/ml if not otherwise stated) were equilibrated in 20 mM phosphate buffer, pH 7.4.

	$[\theta]_{222}$	α-Helix
	deg·cm²·dmol⁻¹	%
ApoA-I		
Low concentration	−15278 ± 145	43
High concentration (1.3 mg/ml)	−21550 ± 345	63
+ TFE[a] 50%	−23574 ± 294	70
+ DMPC	−24540	73
+ NaCl 50 mM	−16191	46
+ NaCl 400 mM	−18819	54
A-I$_M$/A-I$_M$		
Low concentration	−22240 ± 402	66
High concentration (1.3 mg/ml)	−26380 ± 133	79
+ TFE 50%	−25263 ± 272	76
+ DMPC	−25670	77
+ NaCl 50 mM	−22804	67
+ NaCl 400 mM	−21915	65
rA-I$_M$/A-I$_M$		
Low concentration	−21675 ± 615	64
High concentration (1.3 mg/ml)	−24300 ± 10	72
+ TFE 50%	−26150 ± 330	78
+ DMPC	−26450	79
+ NaCl 50 mM	−21596	64
+ NaCl 400 mM	−21925	65

[a] TFE, trifluoroethanol.

α-helical content of ~75% being close to the maximum helical potential deduced from the apoA-I primary sequence (9).

The higher α-helical content of A-I$_M$/A-I$_M$ *versus* apoA-I in water may result from protein-protein interactions between the α-helices of A-I$_M$, brought in close proximity by the disulfide bond, *i.e.* in a way not different from that of self-associated apoA-I (46). The CD spectrum of apoA-I was indeed significantly affected by protein concentrations over the 0.1–1.3 mg/ml range (Table I), as reported by others (35, 46); at low protein concentrations, where the monomer species is predominant, apoA-I displayed a 43% α-helical content, which rose up to 63% when the protein concentration was increased into the region where self-association is significant. Notably, the α-helical content of A-I$_M$/A-I$_M$ was less influenced by protein concentration (Table I), confirming the lower tendency of A-I$_M$/A-I$_M$ to self-associate deduced from cross-linking studies. The α-helical content of self-associated apoA-I was almost identical to that of covalently linked A-I$_M$/A-I$_M$ (64–66%), suggesting that a similar conformational change occurs in apoA-I when two polypeptide chains are brought in close proximity, either by increasing the protein concentration or by introducing a covalent bond.

To monitor the influence of lipid on apolipoprotein conformation, apoA-I, A-I$_M$/A-I$_M$, and rA-I$_M$/A-I$_M$ were mixed with DMPC at 25 °C. The co-incubation resulted in a complete clearing of the turbid lipid dispersion, indicating dissolution of the large phospholipid multilamellar liposomes and the formation of stable lipid-protein complexes (27). Binding to DMPC caused an increase in the α-helical content of all proteins (Fig. 6, Table I). The induction of α-helix in A-I$_M$/A-I$_M$ was significantly lower compared to apoA-I, bringing the spectra of lipid-bound apoA-I and A-I$_M$/A-I$_M$ in close proximity (Fig. 6). All together, the far-UV CD data demonstrate that a maximum α-helical content is reached with both apoA-I and A-I$_M$/A-I$_M$ upon self-association, lipid-binding, or exposure to trifluoroethanol (46, 47, 48).

In other studies, we investigated the effect of solvent pH on the stability of lipid-free apoA-I and A-I$_M$/A-I$_M$ solutions. Plots of $[\theta]_{222}$ as a function of pH are shown in Fig. 7. The ellipticity of both proteins markedly decreased at pH values below 4 and above 10. ApoA-I and A-I$_M$/A-I$_M$ were insoluble at pH between 4.5 and 5.5, and 3.5 and 5.5, respectively. Similar pH-dependent changes in the secondary structure of chicken and human apoA-I have been previously reported (42, 48). Deprotonation of lysine residues and protonation of glutamates and aspartates at extreme pH values causes the disruption of intra- and inter-helix ion pair and charge-dipole interactions, as well as charge repulsions, resulting in the loss of secondary structure. It should be noted that A-I$_M$/A-I$_M$ constantly displayed a significantly higher $[\theta]_{222}$ than apoA-I, over the entire pH range (Fig. 7), the pH-induced decrease in secondary structure being similar to that of apoA-I. It can be argued that ionic interactions are only partially responsible for the increased secondary structure of A-I$_M$/A-I$_M$ *versus* apoA-I, hydrophobic forces being possibly involved in the stabilization of the A-I$_M$ dimer. This conclusion is supported by the observation of a different effect of solvent ionic strength on apoA-I and A-I$_M$/A-I$_M$ secondary structure (Table I). While the α-helical content of A-I$_M$/A-I$_M$ was largely unaffected by ionic strength, apoA-I showed a 3 and 11% increase in α-helical content upon addition of 50 and 400 mM NaCl. Thus, while an increase in the polarity of the medium strength affected the hydrophobic interactions between the amphipathic helical segments of apoA-I, favoring protein self-association (46), a similar structuring effect was not found with A-I$_M$/A-I$_M$. This observation is suggestive that hydrophobic interactions are instrumental in the higher secondary structure of A-I$_M$/A-I$_M$ *versus* apoA-I.

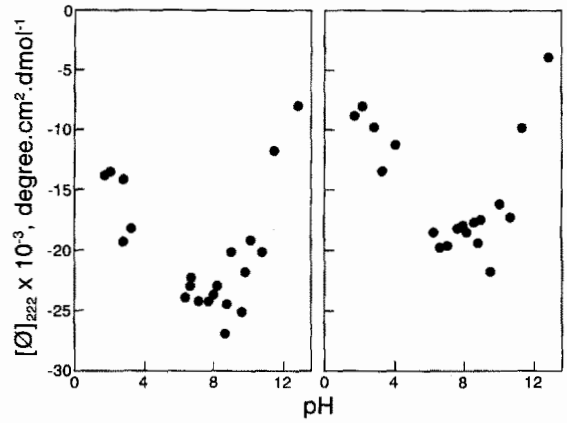


FIG. 7. **Effect of pH on molar ellipticity at 222 nm of rA-I$_M$/A-I$_M$ (*left panel*) and of apoA-I (*right panel*).** Protein samples were adjusted with respect to pH over the range 1.5–12.8 and CD spectra recorded at 25 °C. The final protein concentration was 0.1 mg/ml.

The tertiary structure of lipid-free A-I$_M$/A-I$_M$ and rA-I$_M$/A-I$_M$ was studied by near-ultraviolet CD spectroscopy (Fig. 8). The overall features of the two spectra were almost coincident, and remarkably different from that of apoA-I. The A-I$_M$/A-I$_M$ spectra displayed a broad negative band around 270 nm, extending throughout the near-ultraviolet region, together with signals at 296.8, 291.0, 283.5, 264.0, and 258.0 nm. The 296.8 and 291.0 nm bands can be attributed to 0–0 ^1L_a and 0–0 ^1L_b tryptophanyl transitions, respectively (49), while the band at 283.5 nm may be due to overlapping of tryptophan 0–850 ^1L_b and tyrosine 0–0 ^1L_b transitions. The broad negative band, beginning around 310.0 nm and centered at 270.0 nm, is probably due to the disulfide bond and may overlap the phenylalanyl 0–0 ^1L_a band. The shoulder at 264 nm is assigned to phenylalanyl residues (49). The shape of the A-I$_M$/A-I$_M$ spectra indicates that the aromatic residues are immobilized in a highly asymmetric, hydrophobic environment. The association of A-I$_M$/A-I$_M$ with DMPC did not result in significant changes in the CD spectrum (Fig. 8), indicative of a similar conformation of A-I$_M$/A-I$_M$ in the lipid-free and lipid-bound states. The spectrum of A-I$_M$/A-I$_M$ is drastically different, especially in the 270–300 nm region, from that of lipid-free apoA-I, where a negative extremum at 292 nm

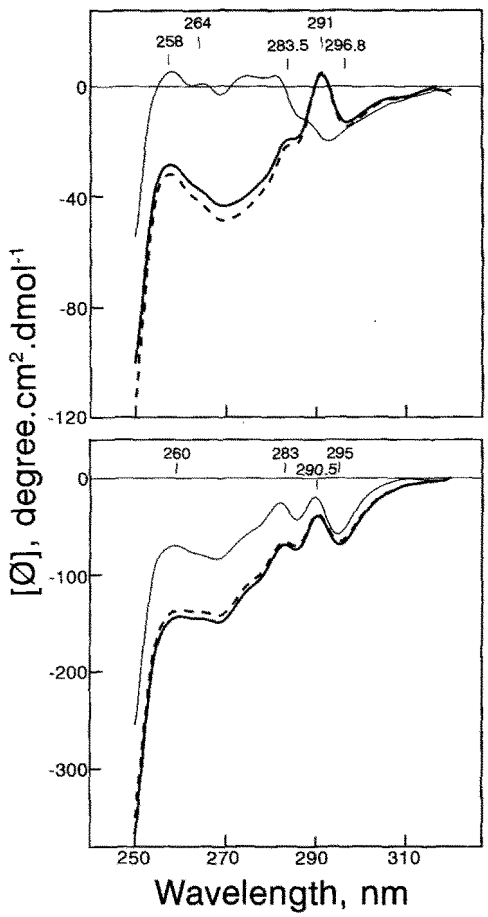


FIG. 8. **Near-ultraviolet circular dichroism spectra of native A-I$_M$/A-I$_M$ (*continuous lines*), rA-I$_M$/A-I$_M$ (*dashed lines*) and apoA-I (*thin continuous lines*), in the lipid-free (*top panel*) and lipid-associated (*bottom panel*) states.** Protein samples (1.0 mg/ml) were equilibrated in 20 mM phosphate buffer, pH 7.4, before spectra were recorded at 25 °C.

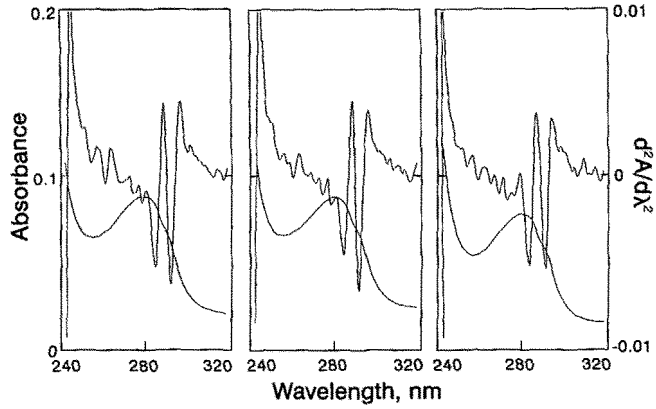


FIG. 9. **Ultraviolet absorbance and second-derivative spectra of rA-I$_M$/A-I$_M$ in 20 mM phosphate buffer, pH 7.4 (*left panel*), in the same buffer containing 6 M GdnHCl (*middle*), and when associated with DMPC (*right*).** Protein concentration was 0.1 mg/ml in all tested conditions. The peak *r* ratio was calculated from the second-derivative spectra as *a*/*b*, where *a* and *b* are the peak to trough distances between the minima at 283 and 290.5 nm and the maxima at 287 and 295 nm, respectively. Fractional tyrosine exposure was then calculated as under "Materials and Methods."

and a shoulder at approximately 286 nm are detectable (Fig. 8). It is of interest that these bands are reversed in sign and shifted to 283 and 290 nm following association of apoA-I with DMPC (Fig. 8). The almost identical shapes of the tryptophan and tyrosine bands in A-I$_M$/A-I$_M$ and lipid-associated apoA-I indicate that comparable changes in apoA-I conformation are induced either by protein-protein interactions between two apoA-I chains, brought in close proximity by a disulfide bond as in A-I$_M$/A-I$_M$, or by association with amphiphile (46, 47, 50), *i.e.* in good agreement with the far-ultraviolet CD data.

Spectroscopic Studies—Tyrosine exposure in apoA-I, A-I$_M$/A-I$_M$, and rA-I$_M$/A-I$_M$ was determined by second-derivative ultraviolet spectroscopy, according to Ragone *et al.* (32). The absorption and second-derivative spectra recorded for the two A-I$_M$ dimers are, again, almost identical; those of rA-I$_M$/A-I$_M$ are shown in Fig. 9. In all tested conditions, *i.e.* proteins in the lipid-free and lipid-associated states or denatured in 6 M GdnHCl, the ultraviolet-absorption spectra revealed absorption maxima at 280 nm with a shoulder at 290.2 nm. The second-derivative spectra demonstrated characteristic peaks at 287 and 295 nm with accompanying troughs at 283 and 290.5 nm (Fig. 9). The peak *r* ratios calculated for apoA-I were 1.10 in phosphate buffer, 0.74 upon association with DMPC and 1.34 in 6 M GdnHCl. These values correspond to a fractional tyrosine exposure (α) of 0.88 for the native protein, 0.46 for the lipid-bound protein and 1.10 for the unfolded state, or exposure of 6, 3, and 7 tyrosine residues, respectively.

The peak *r* ratio determined for A-I$_M$/A-I$_M$ and rA-I$_M$/A-I$_M$ in the denatured state, was of 1.07 and 1.01, respectively. These values are significantly lower than expected from the Tyr/Trp ratio in the protein sequence, and indicate that two out of the 14 tyrosine residues of A-I$_M$/A-I$_M$ are buried in the denatured proteins. The peak *r* ratio decreased to 0.88 and 0.87 in native proteins; association with DMPC did not result in further changes in the peak *r* ratio (0.82 and 0.85). These values correspond to a fractional tyrosine exposure (α) of ~0.60 for native and lipid-bound A-I$_M$/A-I$_M$, or exposure of 8.2 tyrosine residues. This could be an overestimate of the actual number of tyrosine residues which are exposed in the native A-I$_M$/A-I$_M$ molecule, because of the sensitivity of the second-derivative absorption spectrum to changes in the electronic state of buried tyrosine residues (32).

Altogether, the near-UV CD and the second-derivative UV data demonstrate that in A-I$_M$/A-I$_M$ the two A-I$_M$ monomers tightly fold upon themselves, mainly through apposition of the hydrophobic sites of the amphipathic helical segments. Such a folded structure would drastically reduce the flexibility of the A-I$_M$ monomers thus severely affecting their functional properties.

Implications for the Functional Morphology of A-I$_M$/A-I$_M$—The present data indicate that A-I$_M$/A-I$_M$ possesses molecular properties that are unique compared to those of normal apoA-I and monomeric A-I$_M$. The Arg^{173} → Cys substitution in A-I$_M$ occurs at the surface of the molecule, in the middle of one of the amphipathic helical segments of apoA-I (27). By analogy with the x-ray crystal structure of the NH$_2$-terminal domain of human apoE (45), and according to the amphipathic helix theory of apolipoprotein structure (9), the guanidinium group of Arg^{173} in apoA-I is salt-bridged with the oppositely charged carboxylate group of Glu^{169}. The loss of the salt bridge in monomeric A-I$_M$ (27) results in a lower α-helical content and increased flexibility in the interaction of the apolipoprotein with lipids (27). By contrast, the present findings indicate that the introduction of a disulfide bridge in A-I$_M$/A-I$_M$ results in facilitated interhelix interactions, with an increased secondary structure, and a more folded tertiary structure.

The present concept of apoA-I is that of a highly flexible protein, which exists in an equilibrium between a lipid-free form and a bound lipoprotein form. By analogy with the molecular structure of other apolipoproteins (44, 45), lipid-free

apoA-I should form up and down bundles of amphipathic helices. Following binding to lipids, apoA-I spreads on the lipid surface (15, 17), the hydrophobic interactions in the interior of the lipid-free protein being substituted by interactions with lipids. In view of these unique properties, it is not surprising that the introduction of the disulfide bridge in A-I$_M$/A-I$_M$ can result in major changes in molecular organization.

While the definition of the three-dimensional structure of A-I$_M$/A-I$_M$ awaits the availability of quality crystals for structure determination by x-ray crystallographic methods, the present data suggest that the disulfide bridge in the middle of an amphipathic lipid-binding helix would drastically affect the ability of apoA-I to readily convert from a lipid-free to a lipid-bound state, and vice versa. The kinetics of association with lipids, and the ability to desorb from lipoproteins would be remarkably affected. While lipid binding investigations are presently being carried out in this laboratory,[3] previous studies have shown that A-I$_M$/A-I$_M$ confers to carriers' HDL an increased *in vitro* resistance against modifications induced by the interaction with other lipoproteins, enzymes and lipid-transfer proteins (25). The delayed *in vivo* catabolism of A-I$_M$/A-I$_M$ compared to monomeric and normal apoA-I is also indicative of a tighter binding of A-I$_M$/A-I$_M$ to lipids, lipid-free apoA-I being catabolized at a faster rate than in a lipid-bound form (26).

The low degree of structural adaptability of A-I$_M$/A-I$_M$ would also affect the physical properties of stable lipid-protein complexes. Indeed, a careful characterization of plasma lipoproteins in the carriers has clearly shown the presence of small, A-I$_M$/A-I$_M$-containing HDL, which are absent in normal human plasma (51). On the other hand, the peculiar molecular characteristics of A-I$_M$/A-I$_M$ may confer to this protein properties not shared by normal apoA-I; indeed, A-I$_M$/A-I$_M$ is capable to activate plasminogen, by interacting with binding sites on the plasminogen molecule, which apoA-I can not do (52). Finally, the availability of large amounts of recombinant A-I$_M$/A-I$_M$, structurally identical to the native protein, will allow testing *in vivo* whether the unique molecular properties of A-I$_M$/A-I$_M$ result in an improved activity against atherosclerosis development (53).

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